

# Xinyuan LIANG

HYDROLOGY & WATER RESOURCES ENGINEERING

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## Summary

Motivated undergraduate student in Hydrology and Water Resources Engineering (Top 3%). Equipped with substantial work experience in environmental research and ESG consulting, combining strong research capabilities, engineering skills, and creative problem-solving. An effective communicator and collaborator, dedicated to promoting sustainability and environmental protection.

## Professional Experience

|                     |                                                                                                                             |                          |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------|--------------------------|
| Mar 2025 - Jun 2025 | <b>Consulting Intern, Zerenyun (Beijing) Holding Group Co., Ltd.</b>                                                        | <b>Shanghai (Remote)</b> |
|                     | Orchestrated ESG report and framework development for 10+ state-owned enterprises across diverse sectors.                   |                          |
| Sep 2024 - Feb 2025 | <b>ESG Consulting Assistant, Shanghai Boxu Mingguan Technology Co., Ltd.</b>                                                | <b>Shanghai</b>          |
|                     | Analyzed data for a carbon footprint tool to evaluate urban road lighting systems.                                          |                          |
| Jul 2024 - Aug 2024 | <b>Environmental Project Intern, Chinese Research Academy of Environmental Sciences</b>                                     | <b>Beijing</b>           |
|                     | Processed 2022-2023 surface water data (13 indicators, 3000+ points/station) for 88 stations across 11 cities using Python. |                          |

## Field Practices

|          |                                                                               |                    |
|----------|-------------------------------------------------------------------------------|--------------------|
| Jun 2025 | <b>Comprehensive Groundwater Practice</b>                                     | <b>Daqing</b>      |
|          | Conducted hydrogeological mapping and prepared water table contour maps.      |                    |
| Jun 2024 | <b>Hydrological Station Practice</b>                                          | <b>Yichun</b>      |
|          | Operated data measurement instruments in field hydrology.                     |                    |
| Sep 2023 | <b>Engineering Surveying Practice</b>                                         | <b>Harbin</b>      |
|          | Surveyed 20+ sites with a total station and created CAD-based elevation maps. |                    |
| Jun 2023 | <b>Geological Practice in the Liujiang Basin</b>                              | <b>Qinhuangdao</b> |
|          | Documented 10+ types of rock formations through field observation.            |                    |

## Publications

|          |                                                                                                                                         |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------|
| Apr 2025 | Co-author, "Understory Vegetation Regulated the Soil Stoichiometry in Cold-Temperate Larch Forests", <i>Plants-Basel</i> (JCR Q1, IF=4) |
|----------|-----------------------------------------------------------------------------------------------------------------------------------------|

## Research Experience

|                     |                                                                                                                                                                                   |
|---------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Apr 2025 - May 2025 | <b>Design of a Novel Iron-Based Wastewater Treatment Unit</b>                                                                                                                     |
|                     | Developed a prototype industrial wastewater treatment device with iron recovery functionality.                                                                                    |
| Jun 2024 - Jun 2025 | <b>Spatiotemporal Patterns and Driving Factors of Regional Precipitation Use Efficiency</b>                                                                                       |
|                     | Modeled precipitation use efficiency using GIS and identified environmental driving factors.                                                                                      |
| May 2024 - Jul 2024 | <b>Study on Numerical Simulation of Groundwater</b>                                                                                                                               |
|                     | Built a 3D groundwater model to map flow fields and water level distribution.                                                                                                     |
| Jan 2024 - present  | <b>Spatiotemporal Variability and Driving Factors of Precipitation in the Guangdong-Hong Kong-Macao Greater Bay Area from 1901 to 2020, and Analysis of Extreme Precipitation</b> |

|                     |                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                                     |
|---------------------|-------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------------|
|                     | <ul style="list-style-type: none"> <li>Analyzed trends, cycles, and abrupt changes in 120 years of precipitation data for the Greater Bay Area using the Hurst test, wavelet analysis, and Mann-Kendall test.</li> <li>Used wavelet coherence to study the impact of seven meteorological factors on precipitation, including sunspots and ENSO.</li> <li>Applied the geographic detector model to analyze the influence of driving factors such as topography, vegetation, soil, and human activities on precipitation.</li> </ul> |
| Sep 2023 - May 2024 | <b>Characteristics of Rhizosphere Soil Carbon and Nitrogen in Predominant Tree Species of the Xiaoxing'an Mountains</b><br>Compared rhizosphere vs. non-rhizosphere soil C & N content across four tree species.                                                                                                                                                                                                                                                                                                                    |
| Jun 2023 - Jun 2024 | <b>Analysis of Precipitation Characteristics and Prediction in the Sanjiang Plain</b><br>Utilized OCA, Kendall's Rank Correlation Method, and the Run Test to analyze regional precipitation characteristics.                                                                                                                                                                                                                                                                                                                       |

## Education

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|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|
| Sep 2022 - present | <b>Heilongjiang University</b><br>B.Eng. in Hydrology & Water Resources Engineering<br><i>GPA: 89.1/100   Ranking: Top 3%   IELTS: 7 (R8.5, L7.5, W6.5, S6)</i> |
|--------------------|-----------------------------------------------------------------------------------------------------------------------------------------------------------------|

## Extracurricular Activities

|                     |                                                                           |
|---------------------|---------------------------------------------------------------------------|
| May 2025 - present  | <b>Women's Singles Player</b>   Heilongjiang University Tennis Team       |
| Jun 2024 - Jun 2025 | <b>Head of the Competition Department</b>   Football Association          |
| Jun 2023 - Jun 2024 | <b>Head of HR</b>   International Youth Fellowship                        |
| Sep 2022 - Sep 2023 | <b>Member of the Activities Department</b>   Youth Volunteers Association |
| Sep 2022 - Jun 2023 | <b>Defender</b>   Heilongjiang University Women's Soccer Team             |

## Honors & Awards

|             |                                                                                                                                                         |
|-------------|---------------------------------------------------------------------------------------------------------------------------------------------------------|
| Feb 2024    | <b>S Award</b>   Mathematical Contest in Modeling (MCM) in the United States                                                                            |
| Dec 2023    | <b>Outstanding Student</b>   Heilongjiang Province                                                                                                      |
| Oct 2023    | <b>Outstanding Volunteer</b>   Youth Volunteer Association                                                                                              |
| Sep 2023    | <b>Second Prize</b>   Northeastern China Mathematical Modeling Competition                                                                              |
| Aug 2023    | <b>Silver Award</b>   Heilongjiang Province division of the China International "Internet+" College Student Innovation and Entrepreneurship Competition |
| 2023 - 2024 | <b>First-Class Comprehensive Scholarship</b>   Heilongjiang University (only for the top 5% students every year)                                        |
| 2022 - 2023 |                                                                                                                                                         |

## Skills

|                  |                                                                                                                      |
|------------------|----------------------------------------------------------------------------------------------------------------------|
| Technical Skills | GIS, AutoCAD, Origin, SPSS, GMS, Construction surveying instruments                                                  |
| Certifications   | LEED Green Associate (U.S. Green Building Council), National Level 3 Football Referee (Chinese Football Association) |
| Languages        | Chinese (native), English (fluent), French (DELFB1 in preparation)                                                   |
| Others           | Guzheng (Grade 7), Photography (www.liangxinyuan.world)                                                              |